The DysLex-Probe Stimulus Dataset: Full Report

How it was created, every column explained and justified, and how it was used in the experiments

File: `dyslex_probe_stimuli_clean.csv` · 440 rows · 18 columns · UTF-8, Bangla in Unicode Bengali script
All counts and examples in this report were read directly from the committed CSV.

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1. What the dataset is, in one paragraph

The dataset is a controlled set of 440 linguistic stimuli built to probe whether multilingual language models encode phonological and lexical structure that transfers between English and Bangla. It is perfectly balanced: 220 English items and 220 Bangla items. Every item is tagged with 18 columns of metadata so that a probe can be trained on one property (for example "is this a real word?") while the metadata lets us control for confounds such as word length, consonant clusters, or orthographic transparency. The same single file is fed identically to all four models (XLM-R, BLOOM, mT5, Llama), which is what makes the four-model comparison fair.

2. Design principles (why the dataset looks the way it does)

- **Perfect language balance.** 220 vs 220. Any English-vs-Bangla difference cannot be blamed on unequal sample sizes.
- **Category balance within language.** Every task category is split evenly across the two languages (for example minimal pairs are 60 EN + 60 BN), so each probe sees matched amounts of each language.
- **Theory-driven categories.** The categories map onto known dyslexia error types (sound confusions, decoding non-words, syllable load, rhyme, sentence processing), so a positive probe result is meaningful for screening, not arbitrary.
- **Bangla grounded in real clinical data.** Bangla items target the difficulty patterns documented by Badsha (2024, University of Dhaka), then were checked by a native speaker to remove accidental real words and bad conjuncts.
- **Rich metadata for confound control.** The 18 columns exist precisely so the analysis can match items on length and rule out surface shortcuts. This is what allowed us to discover that most probing tasks were being solved by shortcuts.

3. How the dataset was created (pipeline)

1. **Category design.** Seven phonological/lexical task types plus a cross-lingual concept-matching set were defined from the dyslexia literature.
2. **English items** were written or drawn from standard word lists, with IPA transcriptions and matched

real/pseudo pairs.

3. **Bangla items** were authored to mirror the documented dyslexia patterns (voiced/unvoiced stops, aspiration, conjuncts, matra), stored in Unicode Bengali script.
4. **Native-speaker review** removed Bangla pseudowords that were accidentally real, and corrected conjunct encodings.
5. **Metadata tagging.** Each item received the 18-column annotation (complexity, transparency, clusters, difficulty, targeted error, source, pair links).
6. **Source tagging.** Items taken from the source dissertation are marked in `dissertation_source`; the rest are marked `generated`.
7. **Freeze and commit.** The cleaned file is committed once and reused unchanged across all four model runs for reproducibility.

Source provenance in the committed file: 331 items are `generated`; 109 items are tagged to Badsha "Findings ch.6", broken down as voiced-sound difficulty (30), multisyllabic difficulty (20), rhyming sentences (20), Bangla-specific contrasts (16), and phoneme-grapheme harmonization (13).

4. The 18 columns: full data dictionary

For every column: what it holds, the values actually present in the file, why we collected it, and how it was used.

Column	Values in file	Why we took it	How it was used
`id`	Unique, e.g. `MP_EN_001`, `RW_BN_002`	Stable identifier encoding category + language + number, so any item is traceable.	Joins results back to items; lets us audit exactly which items a probe got right or wrong.
`language`	`english` (220), `bangla` (220)	The core independent variable of the whole study.	Defines the train/test split for cross-lingual transfer: train on English, test on Bangla.
`task_category`	8 values (see §5)	Tells us which phonological/lexical skill each item probes.	Selects the item subset for each probing task (voicing, realword, etc.).
`word`	The stimulus text; Bangla in Bengali script (e.g. পানি, গাল)	The actual input fed to the model.	Tokenized and passed through each frozen model to extract hidden states.
`ipa`	IPA string; empty for the 30 sentences	Records pronunciation independent of spelling, so we can reason about sound vs	Reference for phonological design and for checking the voicing/sound contrasts; sentences intentionally left

		script.	blank.
`syllable_count`	Integer; for sentences = word count	Phonological load increases with syllables; a known dyslexia stressor.	Target of the syllable-count probing task; also a length/complexity control.
`phonological_complexity`	`low` (191), `medium` (175), `high` (74)	Coarse difficulty of the sound structure.	Stratification and analysis covariate; lets us check whether models struggle more on complex items.
`orthographic_transparency`	`transparent` (366), `opaque` (74)	How directly spelling maps to sound; the English/Bangla depth contrast in one column.	Target of the orthographic-regularity probing task.
`voiced_unvoiced`	`na` (410), `voiced` (15), `unvoiced` (15)	Voicing confusion is a documented dyslexia error; labels the minimal-pair contrasts.	Target of the voicing probe. Only the 30 English minimal-pair items carry a real label; see the caveat in §7.
`is_real_word`	`True` (390), `False` (50)	Real-vs-pseudoword decoding is the central dyslexia skill.	Target of the real-vs-pseudoword probe: the one task that survived all sanity checks.
`has_consonant_cluster`	`False` (355), `True` (85)	Clusters (and Bangla conjuncts) are decoding stress points.	Confound control and analysis covariate.
`has_irregular_mapping`	`False` (371), `True` (69)	Irregular spelling-to-sound mappings are harder to decode.	Confound control; supports the orthographic-transparency analysis.
`difficulty`	`easy` (251), `medium` (111), `hard` (78)	Overall human-judged item difficulty.	Stratification and error analysis.
`pair_id`	Non-empty on 160 items (minimal pairs + rhymes)	Links the two members of a within-language pair (e.g. bat/pat).	Keeps paired items together so a pair is never split across train and test.
`matched_pair_id`	Non-empty on 55 items (e.g. `XLING_001`)	Links a concept across languages (e.g. water ↔ পানি).	Defines the cross-lingual concept-matching items and supports transfer analysis.
`dissertation_source`	`generated` (331) or a Badsha ch.6 tag (109)	Records provenance and clinical grounding.	Lets us separate clinically-sourced items from generated ones during review.

`error_type_targeted`	11 values (see §6)	Names the exact dyslexia error each item probes.	Connects each item to a screening-relevant error pattern.
`notes`	Free text	Human notes during authoring/review.	Audit trail; not used as a model input.

5. The eight task categories (with real examples)

Category	EN / BN	What it probes	Example (EN)	Example (BN)
minimal_pair	60 / 60	One-sound contrasts (voicing, vowel, place)	bat / pat	কাল / গাল (kal/gal)
multisyllabic	30 / 30	Syllable load and breakdown	cat, dog ...	মা, বই ...
pseudoword	25 / 25	Decoding non-words (no lexical lookup)	blip, flonk	নিদা, পেলু
real_word	25 / 25	Real-word recognition (paired with pseudo)	clip, plonk	মাটি, পানি
phoneme_grapheme	25 / 25	Sound-to-letter mapping	cat, dog	কাজ, ভাত
rhyme	20 / 20	Rhyme judgment	cat / hat	চাঁদ / বাদ
concept_match	20 / 20	Cross-lingual concept pairs (translation equivalents)	water, fire	পানি, আগুন
sentence	15 / 15	Sentence-level reading processing	"The cat sits on the mat."	আমি বই পড়ি।

Total 440. Note the real/pseudo and concept_match items carry `matched_pair_id` links so a real word and its matched pseudoword, or an English concept and its Bangla equivalent, are kept aligned.

6. Targeted error types (the clinical link)

The `error_type_targeted` column connects each item to a dyslexia error pattern. Distribution in the file:

Error type	Count	Error type	Count
none (control items)	110	multisyllabic_breakdown	40
voiced_unvoiced_confusion	60	rhyme_judgment	40
pseudoword_decoding	50	vowel_substitution	30
phoneme_grapheme_mapping	50	sentence_processing	30

aspiration_confusion	16	substitution	14

7. The labels we probed, and the critical voicing caveat

Four columns were used as probe targets:

- `is_real_word` → the real-vs-pseudoword task (the survivor).
- `orthographic_transparency` → the orthographic task (failed sanity).
- `syllable_count` → the syllable task (failed sanity).
- `voiced_unvoiced` → the voicing task (failed sanity).

> **Why the voicing column is a trap.** In the committed file, `voiced_unvoiced` is labelled only for the 30 English minimal-pair items; all 410 other rows, including every Bangla item, are `na`. To run a Bangla voicing probe at all, the pipeline derived the Bangla voicing label from rule-based "voiced starter" and "unvoiced starter" character sets, i.e. **from the first character of the word**. That made the label 100% recoverable from the first letter, so the probe was reading letter identity, not phonology. This is the central sanity-check finding: the Bangla voicing column is unusable for phonological probing, and any apparent voicing success reflects first-character lookup.

8. How the dataset was used in the experiments

1. **Representation extraction.** Each of the 440 `word` values is tokenized and passed through each frozen model; hidden states are saved at every layer.
2. **Probing.** A linear probe is trained to predict a label column (e.g. `is_real_word`) from those hidden states.
3. **Cross-lingual protocol.** The probe is trained on `language == english` rows and tested on `language == bangla` rows; `pair_id` and `matched_pair_id` keep paired items from leaking across the split.
4. **Sanity controls built from the metadata:** a first-character baseline (exposes the voicing trap), a 5-feature surface baseline using length-type fields, and 1:1 length matching of real vs pseudowords.
5. **Length-matched subset.** For all honest RQ2/RQ3 reporting we use a real-vs-pseudoword subset of n = 50 per language, matched 1:1 on character length (mean chars EN 6.88, BN 5.20), deterministic at `random_state = 42`, committed as `results_realword_length_matched.csv`.
6. **Causal ablation.** On that subset, top-ranked units are zeroed to test where the cross-lingual signal lives (the mT5 localization result).

> **One-line summary of usage:** the `word` column is the model input; the four label columns are probe targets; and the remaining metadata columns exist to control confounds, keep pairs aligned, and build the sanity baselines that ultimately invalidated three of the four tasks.

9. Known issue: CSV vs thesis appendix mismatch (please reconcile)

> The committed CSV and the thesis appendix table (`tab:stim-distribution`) do not list the same categories, although both total 440.

- **CSV (actual data):** 8 categories, including `concept_match` = 40 (20 EN + 20 BN), and no separate "extended" sets.

- **Appendix table:** 7 categories plus "Real word (extended)" 20 and "Pseudoword (extended)" 20; no `concept_match` row; and it mentions "15 cross-lingual concept pairs" whereas the CSV has 40 concept_match items and 55 rows carrying a `matched_pair_id`.

> The base seven categories and their counts match exactly between the two; only the final 40 items are described differently. Recommend updating the appendix table to add a `concept_match` (cross-lingual) row and remove the "extended" rows, so the thesis matches the data file you actually ran. This is worth fixing before the viva, since an examiner comparing the appendix to the CSV would notice.

10. Likely dataset questions in the viva

Why 440 items and not more?

It is large enough to balance 8 categories across 2 languages with matched pairs, while staying small enough for every item to be native-verified and individually justified. We trade scale for control and clinical grounding.

How do you guarantee the comparison across models is fair?

The identical frozen CSV is fed to all four models; nothing is per-model. Differences therefore come from the models, not the data.

Why so much metadata per item?

Because confound control is the whole point. The length, cluster, transparency and pair columns are what let us match items and expose the surface shortcuts. Without them we could not have caught the voicing and length confounds.

Is the dataset biased toward easy items?

Difficulty skews easy (251 easy, 111 medium, 78 hard), which we report. We mitigate by analysing within difficulty and by focusing conclusions on the length-matched subset rather than raw accuracy.

Why keep the unusable voicing column at all?

Transparency. Leaving it in, and reporting that it is first-character recoverable, is exactly how we demonstrate the sanity-check lesson. Deleting it would hide the most important methodological finding.

Every count, value distribution, and example in this report was read directly from `dyslex_probe_stimuli_clean.csv`. Where the CSV and the thesis appendix disagree, the CSV is treated as the ground truth and the disagreement is flagged in §9.